

Imanol Gómez

Technical lead & AI engineer

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Profile

Technical lead and AI engineer with 15+ years of building production software end to end in Python and C++, from automotive AI and medical devices to interactive installations. As Head of AI Deployment at CARIAD (Volkswagen Group), I lead 12 engineers taking AI models from research prototypes to production on a volume-production ADAS platform, working directly with Volkswagen Group brands and with Qualcomm, Samsung and NVIDIA. Before that I built an AI tele-monitoring medical product from requirements to production under IEC 60601, and as a creative technologist I delivered installations, taught, gave talks and ran workshops. In my own time I build LLM agents with automated evaluation and on-device AI on NVIDIA Jetson.

Skills

LLM & agents	Agentic workflows (LangGraph), RAG & pgvector, MCP, Structured outputs (Pydantic), Prompt engineering, LiteLLM / OpenRouter
Machine learning	PyTorch, TensorFlow, Fine-tuning, ONNX, TensorRT, Vision-language driving planner (EMMA-VLM), Model optimization & deployment
Software engineering	Python (production), C++, Rust, FastAPI, PostgreSQL, Docker & Compose, GitHub Actions / CI-CD, HTMX / Jinja, Linux & snap packaging
Edge & embedded	Qualcomm NPU, NVIDIA Jetson Orin, ROS 2, Embedded Linux / QNX, RTOS & MCUs, Firmware, GLSL, CUDA (Particle Atlas, open source)
Regulated environments	IEC 60601 (healthcare), ISO 26262 (automotive), ASPICE, Compliance & auditability
Delivery & leadership	Team leadership (12 engineers), Product ownership (SAFe PO/PM), Stakeholder management, Mentoring, Talks & workshops

Professional experience

Oct 2025 – Present
Berlin

CARIAD

Head of AI Deployment Case study

Leads a team of 12 engineers that takes AI models from research prototypes to production on a volume-production ADAS platform (CARIAD–Bosch Automated Driving Alliance).

- Owns production reliability, performance and auditability under ISO 26262 and ASPICE: latency, memory and power targets, release quality and traceability.
- Works directly with enterprise stakeholders at Volkswagen Group brands (Volkswagen, Audi) on requirements, priorities and delivery.
- Leads technical discussions with semiconductor partners Qualcomm, Samsung and NVIDIA on integration and performance.

Apr 2023 – Oct 2025
Berlin

CARIAD

Senior ML Software Engineer Case study

Technical Product Owner for the AI Runtime Environment; developed and deployed AI perception models on Qualcomm accelerators.

- Built a Python SDK for model conversion, quantization and optimization, plus a runtime abstraction layer used by engineering teams across the organization.
- Supported internal customer teams end to end, from scoping their use cases to production integration.
- Developed sensor-fusion, 3D-occupancy and behaviour-prediction models and deployed them on Qualcomm AI accelerators under real-time constraints.

Aug 2020 – Apr 2023
Berlin

NELA by Carepath Technologies
AI Software Engineer → Senior IoT Engineer
Built NELA Cough, an AI-based tele-monitoring product, in a regulated healthcare environment.

- AI model design and real-time deployment on edge devices and MCUs, in line with IEC 60601.
- Led the IoT system architecture end to end, from requirements to production: Azure IoT cloud back end and device firmware.
- Worked with external medical hardware suppliers through to series production.

Jun 2014 – Aug 2020
Berlin & international

Self-employed
Creative Technologist Case study
Scoped and delivered client projects end to end: interactive installations, light and video projection, live performance software and fashion tech.

- ProtoPixel (2017 – 2020): firmware and software for real-time LED lighting products; delivered the main product software as a snap package; technical evangelist.
- ElektroCouture (2016 – 2018): led design and development of wearable products; main developer of the wearable driver firmware; fashion-tech mentor.
- Engineered and ran live the show software for the multimedia dance performance Murmur (Aakash Odedra Company) on its world tour.

Oct 2011 – Jul 2014
Linz

Ars Electronica Linz
Creative Engineer
Built interactive computer-vision installations in C++, Python, OpenGL and OpenCV.

- Lead developer of GeoPulse, a spatial-data visualization platform (Ars Electronica Center).
- Techkriti 2012 "Brain Battle", MED-EL Audioversum exhibits.

Projects

- **Alibi: LLM agents that lie** github.com/ImanolGo/alibi
  Python, LangGraph, pgvector, MCP, FastAPI, LiteLLM, OpenTelemetry
  A murder-mystery game where every suspect is an LLM agent — with RAG memory, tool use and eval gates in CI. 2.9% contradiction rate across evaluation runs, at $0.03 per run.
- **A fully on-device AI robot assistant** github.com/ImanolGo/rover-assistant
  Python, ROS 2, TensorRT, DeepStream, Whisper, YOLO, llama.cpp
  Voice and vision on an NVIDIA Jetson Orin Nano — nothing leaves the robot. Vision answers in ~37 ms and the language model runs at ~30 tokens/s, all on the robot.
- **EMMA-VLM: driving plans from vision models** github.com/ImanolGo/emma-vlm
  Python, OpenRouter, vision-language models, nuImages, nuScenes
  An EMMA-style driving planner with interchangeable vision models and trajectory evaluation against recorded driving data.

Open source: contributor to openFrameworks (C++); 80+ public repositories on GitHub.

Talks & workshops: Llum BCN Festival · Factory Berlin · Porto Tech Hub · iSpace Accra · Goethe-Institut Bangkok · Scope Sessions Berlin; university teaching at Universitat Pompeu Fabra and HfG Offenbach.

Education

2009 – 2011
Barcelona

Master's Degree, Sound and Music Computing
Universitat Pompeu Fabra
Focus: computational modeling, perception, cognition and interactive systems.

2003 – 2009
Madrid

Master's Degree, Telecommunication Engineering
Universidad Autónoma de Madrid
Specialization: Multimedia Signal Processing, Electronic Systems.

- Erasmus year at TU Berlin (2007 – 2008).
- Research student at Fraunhofer HHI, Image Processing Department (2008 – 2009): forward error correction for video compression.

Languages

Spanish Native **English** Professional proficiency **German** Professional proficiency